

Dhruv Nirmal

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PROFESSIONAL SUMMARY

I work at the intersection of anomaly detection, data engineering, and production AI, turning large, messy operational datasets into clear signals that surface what is wrong and why. My day-to-day is identifying cost-leakage and behavioural patterns in enterprise transaction data, building the dashboards and automated workflows that bring those findings to the people who act on them, and evaluating AI-generated outputs for accuracy before they reach stakeholders. I have also designed and shipped production LLM and multi-agent systems, so I understand how AI tools behave from the inside, not just as a consumer. That combination of behavioural-pattern detection on real data, hands-on AI building, and client-facing consulting delivery is what I bring to GitHub and Copilot.

KEY SKILLS

Analytics & Detection: SQL, Python, Anomaly Detection, Behavioural Pattern Analysis, Statistical Modelling, Data Storytelling, KPI Analytics, Investigative Analysis

BI & Visualisation: Power BI, Tableau, Dashboards, Exception Reporting, Data Visualisation

AI & LLM Systems: LLM Workflows (Claude, ChatGPT), Multi-agent Orchestration, Agentic AI, AI-output Evaluation, Sentiment Analysis, NLP

Data Engineering & Quality: ETL Pipelines, Data Validation, Data Cleaning, Data Quality Controls, PySpark, Databricks, Snowflake, KNIME

Cloud & Infrastructure: Azure, AWS (S3, EC2, IAM), Terraform, Docker

Collaboration & Consulting: Cross-functional Partnership, Stakeholder Communication, Client-Facing Delivery, Technical Communication

EXPERIENCE

Data Analyst, Purchasing Index Data Analytics (Comprara Group)

Jun 2024 to Present | Melbourne, Australia

- Applied anomaly detection to procurement transaction datasets to surface cost-leakage and irregular spending patterns, contributing to roughly 15% client savings by distinguishing normal supplier behaviour from problematic activity.
- Owned the end-to-end analytics pipeline for the largest client account: roughly 6M invoice rows across 13 global sub-regions, with automated ingestion, validation, and an 11 to 12k-rule categorisation engine. Built automated data-quality controls (row counts, column-consistency checks, discrepancy logging) that flag integrity issues before they reach stakeholders.
- Delivered 16+ Tableau dashboards and exception-reporting views across 5 enterprise clients, translating complex spend and supplier data into actionable insights for CFOs, Chief Procurement Officers, and category managers.
- Built a multi-agent AI validation workflow (5 agents plus an orchestrator, Python, Claude) to systematically check client datasets before delivery, and deployed agentic AI workflows for automated client-report generation, saving roughly 12 to 15 hours per week company-wide.
- Developed an LLM-driven call-transcript analyser (placed 1st at an internal AI hackathon) that scores call quality, expectation fulfilment, and sentiment, then routes findings to internal stakeholders via an automated notification pipeline. Adopted into internal operations.

- Built an NLP classification model (TF-IDF, Logistic Regression) to categorise a roughly A\$12B spend dataset, cutting a categorisation cycle from 1 to 1.5 months down to a single model run.
- Designed and ran the client relationship for a restaurant chain across AU and NZ as the sole analyst on the account: structured a 3,000 to 3,500-product catalog, built pricing-decision and supplier dashboards surfacing cost-leakage to the product and venue level, and identified roughly A\$2M in savings over a year.

Data Engineer Intern, Victorian Centre for Data Insights (VCDI)

Aug 2023 to Nov 2023 | Melbourne, Australia

- Built a distributed anomaly-detection pipeline in Databricks (PySpark) to surface irregular patterns in government procurement data, improving detection accuracy by roughly 20%.
- Deployed a Power BI analytics solution adopted by senior Department of Transport stakeholders; presented system architecture and detection outcomes to cross-functional teams.

Research Data Analyst, Terminal Ballistics Research Laboratory (TBRL), DRDO

Jan 2021 to Jul 2021

- Analysed blast-wave signal datasets using statistical models and Butterworth filtering, improving noise-level prediction accuracy by roughly 20% and refining defence-equipment designs.

PROJECTS

Anomaly Detection Pipeline (Databricks, PySpark, Cloud)

- Built a distributed pipeline to detect irregular patterns in large government procurement datasets, engineering scalable transformation layers for cloud processing. Improved detection accuracy roughly 20% versus the prior approach.

LLM Call-Transcript Analyser (Python, LLM Workflows, Agentic AI)

- Designed an agent-orchestrated pipeline that ingests call transcripts, scores quality and sentiment across multiple dimensions, and routes findings to stakeholders via an automated notification pipeline. Placed 1st at an internal AI hackathon; adopted into the company workflow.

Multi-Agent Data QA Validation Workflow (Claude, Python, Agentic AI)

- Built a 5-agent plus orchestrator pipeline to validate client datasets before delivery. Caught data-integrity issues systematically and raised the consistency of analytics outputs across enterprise clients.

Cloud Data Platform with Terraform (AWS, Terraform, Snowflake, Power BI)

- Provisioned full cloud infrastructure as code (S3, IAM, EC2); built an automated API-to-Snowflake ingestion pipeline with a connected Power BI reporting layer; modular multi-environment design (dev, staging, prod).

EDUCATION

Master of Data Science, Monash University | Feb 2022 to Dec 2023

Statistics I & II, Machine Learning, Applied Forecasting, High Dimensional Analysis, Communicating with Data

Bachelor of Engineering, Thapar University | May 2017 to May 2021

Mathematics I/II/III, Statistics, C/C++, Analog & Digital Electronics